higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## INTRODUCTION TO SYSTEMS DEVELOPMENT
**(**Second Paper**)**
**NQF LEVEL 2**

(10041022)

**14 March 2019 (X-Paper)**
**09:00–13:00**

**This question paper consists of 17 pages.**

NCV

Copyright reserved 

**TIME: 4 HOURS**
**MARKS: 100**

---

## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.

---

## INSTRUCTIONS TO STUDENTS AND LECTURERS

1. Students MUST be seated in the computer laboratory 30 minutes before the start of the examination to ensure that all computers have the appropriate programs.
2. These questions must be completed on a computer by making use of MS Office 2003 or a later version and VB.NET 2005 or VB.NET Express or VB.NET 2008 or later.
3. Students must make sure that they print their work immediately after they completed a question. Ensure that your examination number is on ALL the printouts. NO questions WITHOUT a PRINTED EXAMINATION NUMBER will be marked.
4. In the event of a problem such as a power failure, computer breakdown or a printer breakdown, the invigilator will make the necessary arrangements to continue with the examination process once the problem has been resolved. Students will not be penalised for time lost under these circumstances.
5. All work on the computer should be saved at regular intervals to prevent loss of work. NO additional time may be allowed for work lost due to lack of saving.
6. No student may print for another student, or make a memory stick available to another student or access another student's work on the network. Any attempt to access any information from or transfer information to another student in whatever manner is a contravention of the examination rules and regulations and will be viewed in an extremely serious light.

Copyright reserved 

7. All printouts and screen prints done during the session MUST be handed in.

8. Make sure that your EXAMINATION NUMBER appears on ALL printouts.

9. Students may NOT copy the source code from VB.NET to MS Word and then print it from the word processor.

10. STEPS TO PRINT A VB.NET FORM OR ANY OTHER SCREEN:

    10.1 Run the solution.

    10.2 Hold the ALT key down and press the print screen button. (This will copy the active screen.)

    10.3 Open MS Word and paste the form/screen in a new document.

    10.4 Add your EXAMINATION NUMBER to the top right-hand side of the page.

    10.5 Print the page.

11. You are allowed to make use of the help files in VB.NET.

12. The following files will appear on your hard drive in the folders:

    12.1 QUESTION 1
    *… \Lawyers*

    IMAGE: lawyer.png

    *… \IMAGES*

    12.2 QUESTION 2
    *… \BitcoinConverter*

    12.3 QUESTION 3
    *… \TilingCostCalculator*

    12.4 QUESTION 4
    *… \PlayerComparison*

Copyright reserved 

## QUESTION 1

The following questions refer to the VB.NET program called ***Lawyers*** that has been saved on your computer.

A law firm called Spectre & Ross approaches your software development team to help them develop a program which will help them speed up the process of calculating and processing payments made by their clients.

The program allows the user (lawyer) to enter his/her ID, select the case type, enter the client information, select the payment rate and enter the number of hours to be billed.
When the user clicks the display button, payment information is displayed in a label at the bottom of the program.

Your team leader assigns you the job to design the graphical user interface for the program.

**NOTE: NO CODE IS REQUIRED FOR THIS QUESTION. YOU ARE ONLY REQUIRED TO CREATE THE GRAPHICAL USER INTERFACE.**

Use FIGURE 1 below and the instructions which follow as a guide to help you complete the interface.

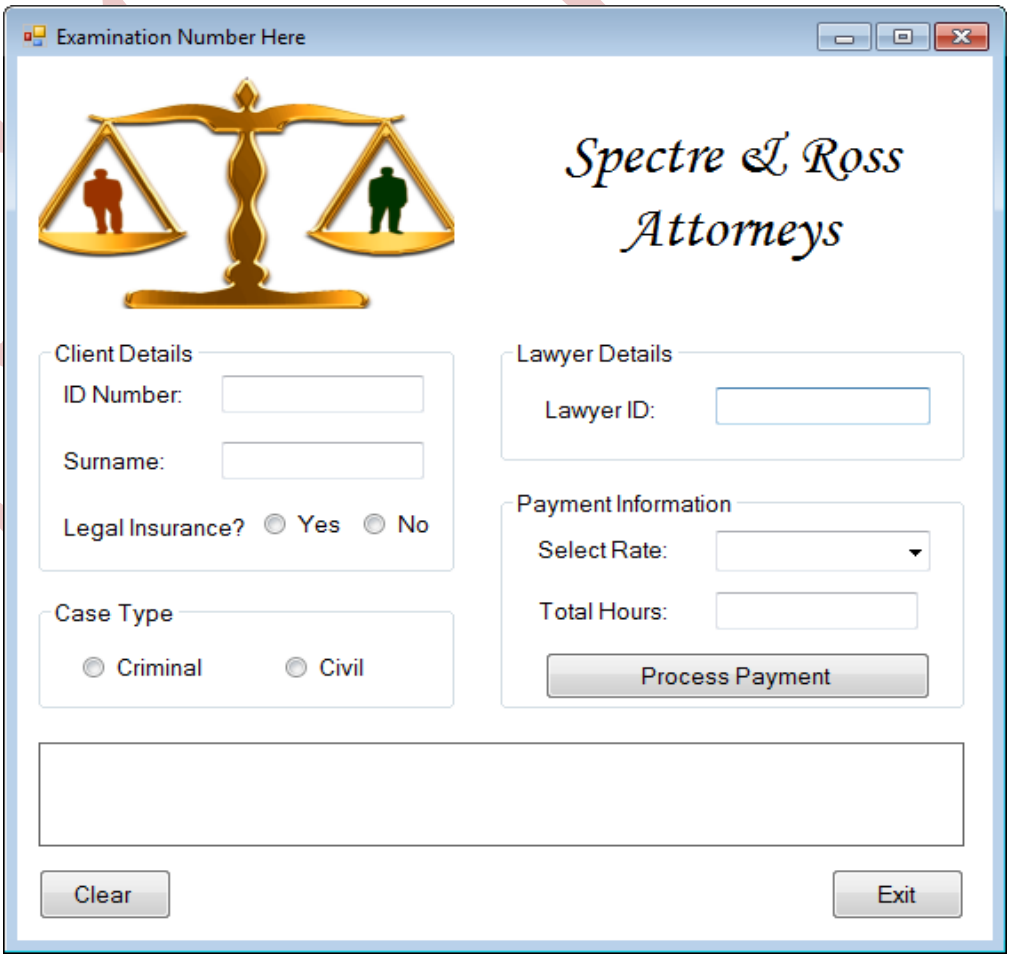



**FIGURE 1**

A blank interface has been saved on your computer. Open the program ***Lawyer*** and follow the instructions below to complete the interface.

1.1 Add a PICTURE BOX to the form as part of the heading:

| CONTROL | NAME | IMAGE |
|---|---|---|
| PictureBox | picLawyer | Use the image saved on your computer as 'lawyer.png' |

(2)

1.2 Add FOUR GROUP BOXES to the form, to group the *Lawyer Information, Client Details*, *Case Types* and *Payment Information*:

| CONTROL | NAME | TEXT |
|---|---|---|
| Group box | grpLawyerDetails | Lawyer Details |
| Group box | grpClientDetails | Client Details |
| Group box | grpCaseType | Case Type |
| Group box | grpPayment | Payment Information |

(3)

1.3 Add EIGHT LABELS to the form, one for the heading, one for the lawyer ID, three for the client details, two for the payment information and one to display the payment information:

| CONTROL | NAME | TEXT | SIZE | FONT |
|---|---|---|---|---|
| Label | lblHeading | Spectre & Ross Attorneys | 26 | Monotype Corsiva |
| Label | lblLawyerID | Lawyer ID: | Default | |
| Label | lblClientID | Client ID: | Default | |
| Label | lblSurname | Surname: | Default | |
| Label | lblInsurance | Legal Insurance? | Default | |
| Label | lblRate | Select Rate: | Default | |
| Label | lblHours | Total Hours: | Default | |
| Label | lblDisplay | (Blank) | **Border:** Fixed Single<br>**Size:** See Figure 1 | |

(10)

1.4 Add FOUR TEXT BOXES to the form to enter the lawyer ID, client ID, client surname and number of hours:

| CONTROL | NAME | TEXT |
|---|---|---|
| Text box | txtLawyerID | (Blank) |
| Text box | txtClientID | (Blank) |
| Text box | txtSurname | (Blank) |
| Text box | txtHours | (Blank) |

(4)

Copyright reserved 

1.5 Add FOUR RADIO BUTTONS to the form to indicate whether or not the client has legal insurance and also to select the case type:

| CONTROL | NAME | TEXT |
|---|---|---|
| Radio button | radYes | Yes |
| Radio button | radNo | No |
| Radio button | radCivil | Civil |
| Radio button | radCriminal | Criminal |

(4)

1.6 Add a combo box to the form to choose the rate:

| CONTROL | NAME | ITEMS |
|---|---|---|
| Combo box | cboRate | Basic, Discount |

(1)

1.7 Add THREE BUTTONS to the form, one to process the payment, one to clear all the fields on the form and one to exit the program:

| CONTROL | NAME | TEXT | FONT SIZE |
|---|---|---|---|
| Button | btnProcess | Process Payment | Default |
| Button | btnClear | Clear | Default |
| Button | btnExit | Exit | Default |

(3)

1.8 Change the text in the title of the form to your EXAMINATION NUMBER and make a printout of the form. (1)

**[28]**

## QUESTION 2

The following questions refer to the VB.NET program called ***BitcoinConverter*** that has been saved on your computer.

As an aspiring Bitcoin trader, you decide to write a program that can help you to quickly convert Bitcoin to Rand and vice versa, in order for you to constantly know how much Rand your current amount of Bitcoin is worth.

The current conversion rate between Bitcoin and Rand is as follows:

1 Bitcoin = R49 820,00

Your program should be able to convert both ways, i.e.:

Bitcoin → Rand AND Rand → Bitcoin

You should be able to select the starting currency, enter the amount to convert and then select the target currency.

When clicking on the *Convert* button the converted amount should be displayed to the user.

Copyright reserved 

Use the IPO chart below to help you with the program:

| INPUT | PROCESSING | OUTPUT |
|---|---|---|
| Select Currency to Convert<br><br>AmountToConvert<br><br>Select Target Currency | Conversion Rates:<br>• 1 Bitcoin = R49 820,00<br>• 1 Rand = 0,00002 Bitcoin<br><br>ConvertedAmount = AmountToConvert * ConversionRate | ConvertedAmount |

FIGURE 2 below is a print screen of what the user interface looks like. The completed interface has been saved on your computer. Open the solution ***BitcoinConverter*** and follow the instructions to complete the program:

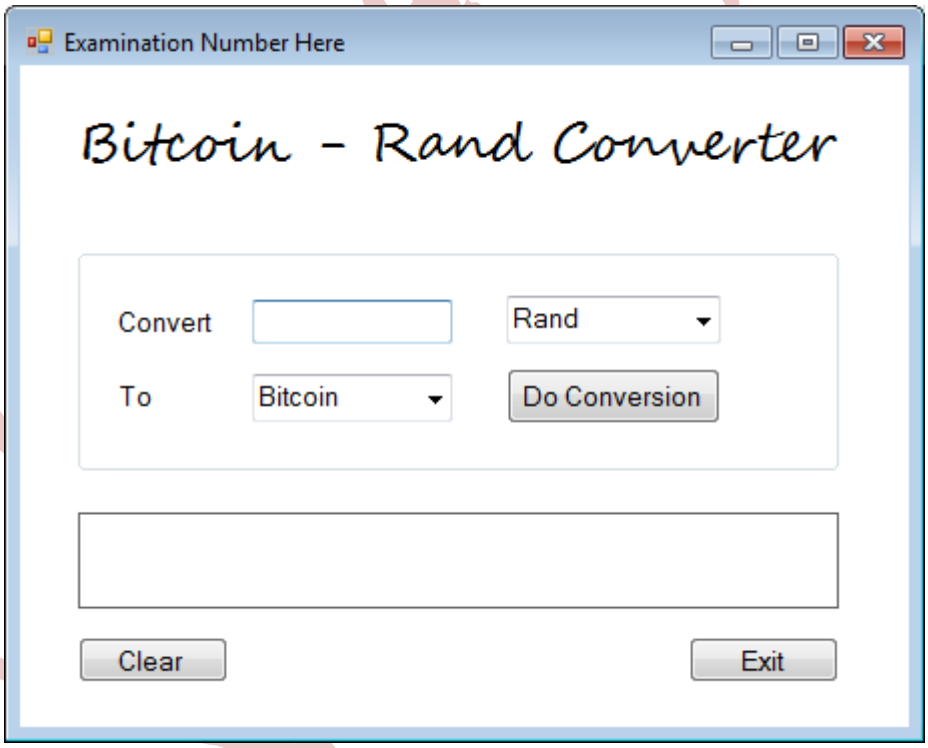



**FIGURE 2**

The following steps explain how to use the program in order to convert Bitcoin to Rand and Rand to Bitcoin:

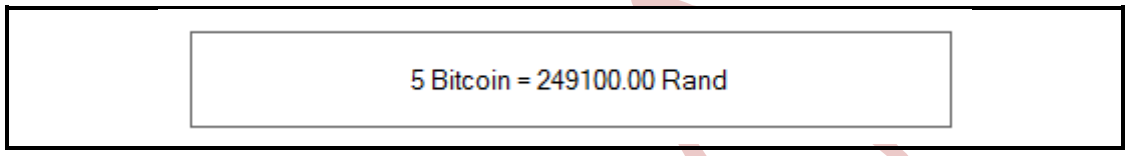
**STEP 1** RUN the program and enter the *Amount to Convert* and select the *Starting Currency:*



**STEP 2** Select the *Desired Currency* and click the *Do Conversion* button:




**STEP 3** The *Converted Amount* is displayed as follows:




2.1 Add the following code to the Form_Load event:

2.1.1 Add the following TWO items to BOTH combo boxes: 'Rand', 'Bitcoin'.

2.1.2 Select the first item in the first combo box (cboConvertFrom) and select the second item in the second combo box (cboConvertTo). (2 × 2) (4)

2.2 Add the following program code to the click event of the *Calculate* button:

2.2.1 Declare ALL the required variables. (3)

2.2.2 Convert and assign the value from the *Amount to Convert* text box to the appropriate variable. (2)

2.2.3 Determine the *Conversion Rate* based on the selections in both the combo boxes using (an) <if> statement(s).

HINT: Make use of the AND logical operator in your *if* statement (6)

2.2.4 Calculate the *Converted Amount.* (3)

2.2.5 Display the calculated *Converted Amount* in the label as shown in Step 3 above. (3)

2.3 Clicking the *Clear* button should clear the textbox and the label, as well as select the first item in the first combo box (cboConvertFrom), and select the second item in the second combo box (cboConvertTo). (4)

2.4 Clicking the *Exit* button should close the program. (1)

Make a printout of the source code.

**[26]**

## QUESTION 3

The following questions refer to the VB.NET program called ***TilingCostCalculator*** that has been saved on your computer.

Your friend does tiling for a living and as such, constantly needs to work out how much each of his tiling jobs will cost his customers. His current method of doing this is by using a pen, paper and a calculator, which is very time-consuming.

He decides to ask you, a software developer, to create a program that will make this task much faster and easier. The program should work as follows:

- The user selects the tile type:
  - Standard Tiles (R15 per square metre)
  - Extra Strength (R20 per square metre)
- The user then enters the length and breadth of the area to be tiled.
- The program then calculates and displays the total cost of the tiling job.

Example: If the area to be tiled has a length of 5 metres and a breadth of 3 metres, the area to be tiled is:

Area = length * breadth
= 3m * 5m
= $15m^2$

If the customer chooses to have normal tiles, the total cost would then be R225. (Normal tiles are R15 per square metre.)

Use the IPO chart below to help you with the program:

| **INPUT** | **PROCESSING** | **OUTPUT** |
|---|---|---|
| Length<br>Breadth<br>Select Tile Type | **Tile Costs**:<br>• Normal Tile: R15 per square metre<br>• Extra Strength: R20 per square metre<br><br>**Area ($m^2$)** = Length * Breadth<br><br>**TotalCost** = Area * Tile Cost | TotalCost |

FIGURE 3 below is a print screen of what the user interface looks like. The completed interface has been saved on your computer. Open the solution ***TileCostCalculator*** and follow the instructions to complete the program:

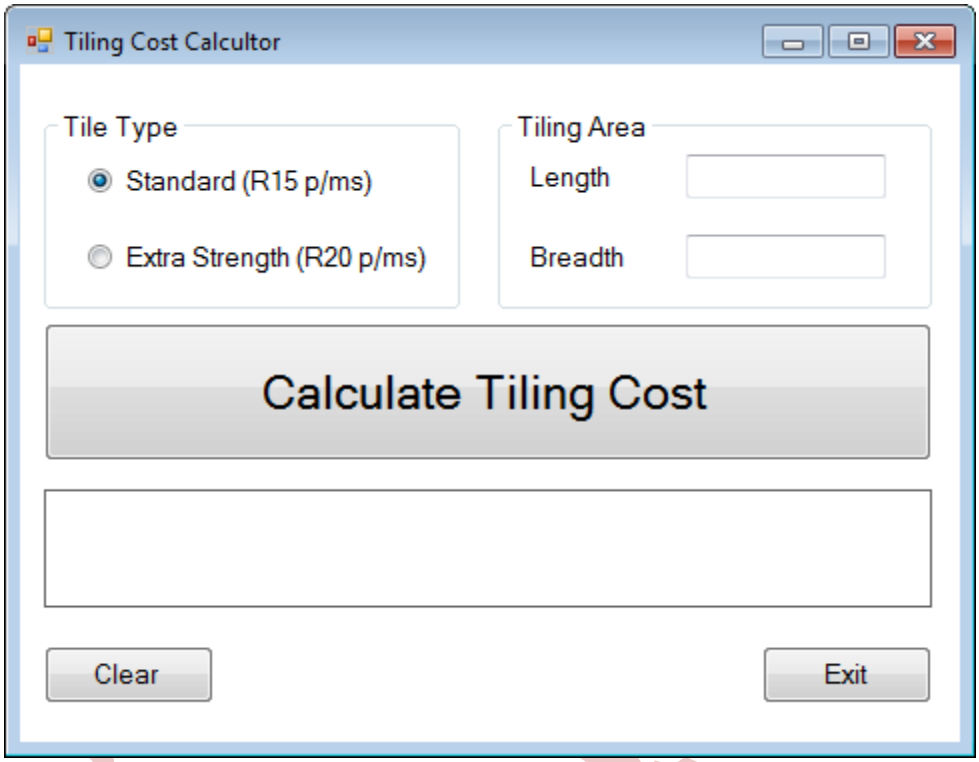


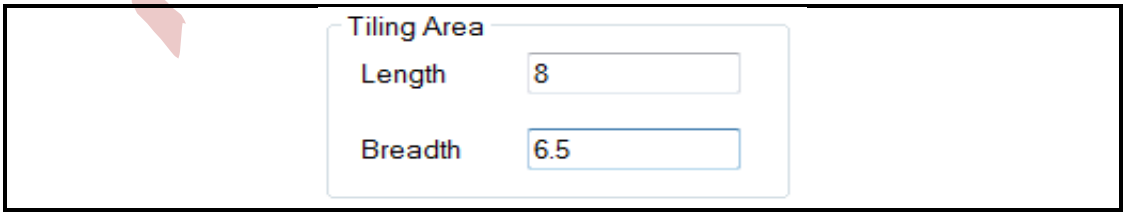

**FIGURE 3**

The following steps explain how to use the program in order to calculate the cost of a tiling job:

**STEP 1** RUN the program and select the *Tile Type*:

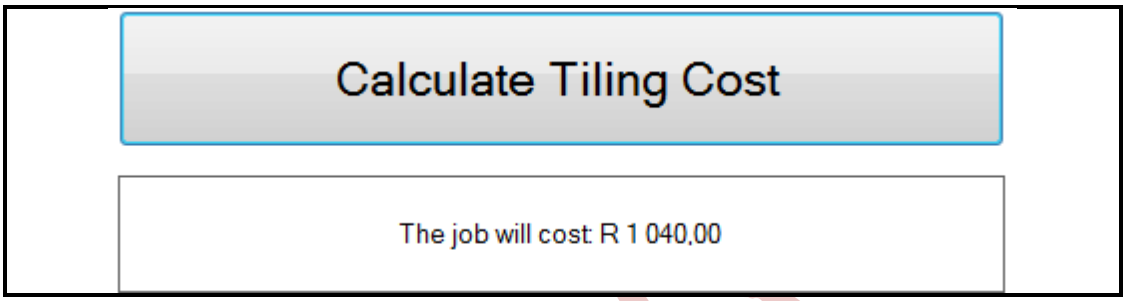



**STEP 2** Enter the *Length* and *Breadth* of the area to be tiled:




Copyright reserved 

**STEP 3** Click the *Calculate* button to calculate and display the *Total Cost* as follows:



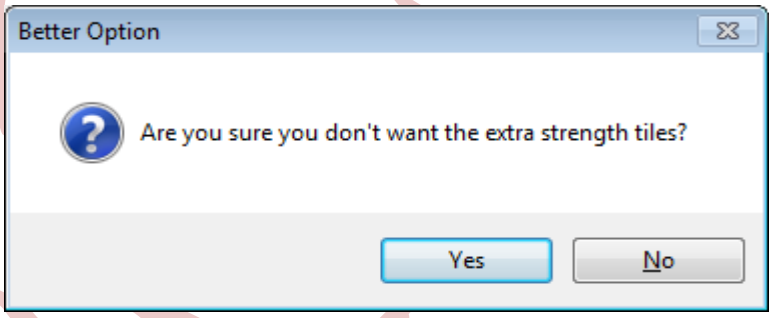

3.1 Add the following program code to the click event of the *Calculate* button:

3.1.1 Write down THREE of the required variables. (3)

3.1.2 Convert and assign the values from the text boxes to the appropriate variables. (2)

3.1.3 Calculate the area. (3)

3.1.4 If *Normal Tile* is selected, the following message should be displayed:




Clicking the *Yes* button should continue with the calculation, while clicking the *No* button should stop the calculation and allow the user to make a different selection. (4)

3.1.5 Determine the *TileCost* based on the selection in the radio buttons by making use of (an) <if> statement(s). (3)

3.1.6 Calculate the *TotalCost.* (3)

3.1.7 Display the calculated *TotalCost* in the label as shown in Step 3 above. (2)

3.2 Clicking the *Clear* button should clear the radio buttons, text boxes and the display label. (3)

3.3 Clicking the *Exit* button should close the program. (1)

Make a printout of the source code.

**[24]**

## QUESTION 4

The following questions refer to the VB.NET program called ***PlayerComparison*** that has been saved on your computer.

Your uncle is the coach of an amateur cricket team and knowing that you are studying computer programming, he asks you to design a program that will help to make it easier for him to select a team.

The program should accept the *name, surname, total number of runs, total number of wickets* and *total games* of two players. It should then compare these two players to each other by calculating their batting (runs) average as well as their bowling (wickets) average. The program should then display which player is the better batsman and which is the better bowler. The averages are calculated as follows:

Batting Average = Total Runs / Total Games
Bowling Average = Total Wickets / Total Games

The player with the higher batting average is then the better batsman, while the player with the higher bowling average is the better bowler.

NCV

FIGURE 4 below is a print screen of what the user interface looks like:

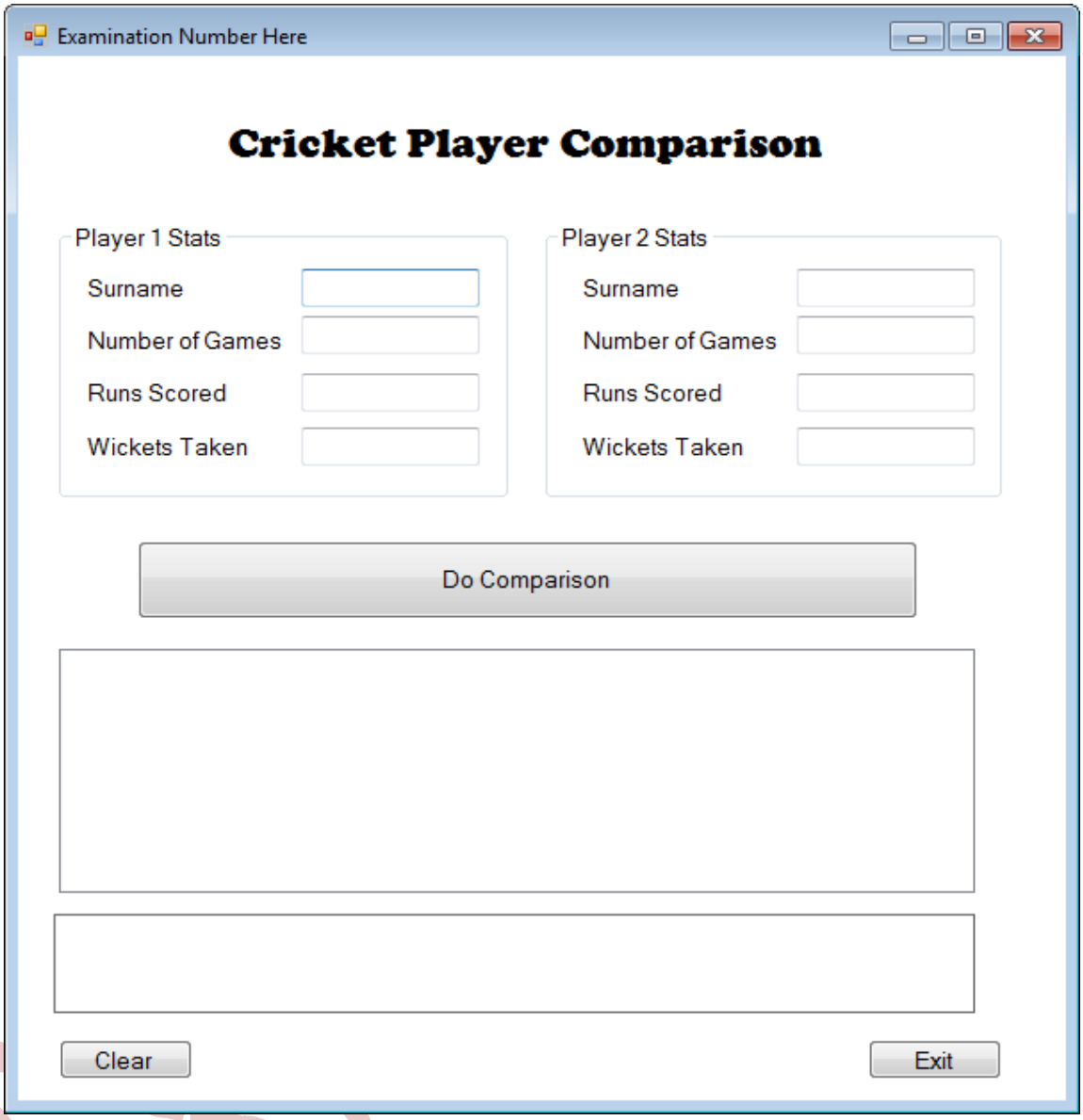
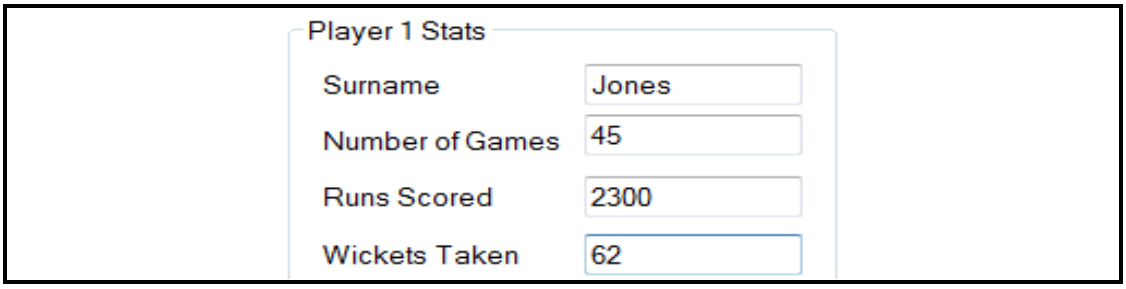



**FIGURE 4**

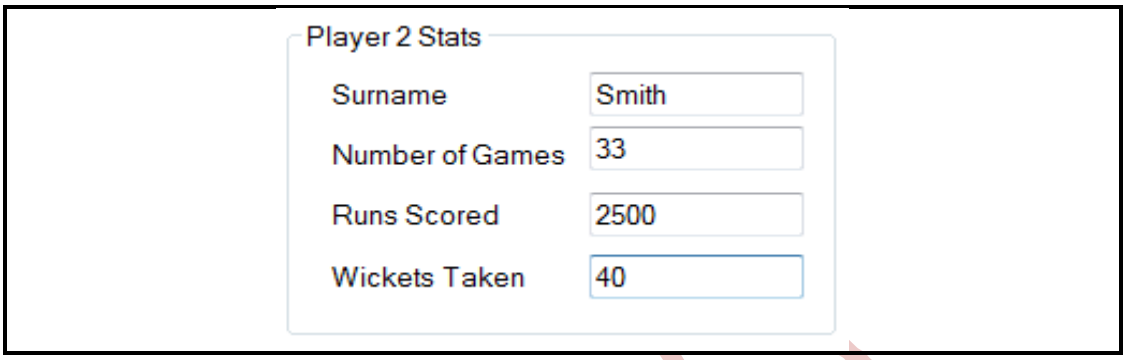
WORKING OF THE PROGRAM EXPLAINED:

The following steps explain how to use the program:

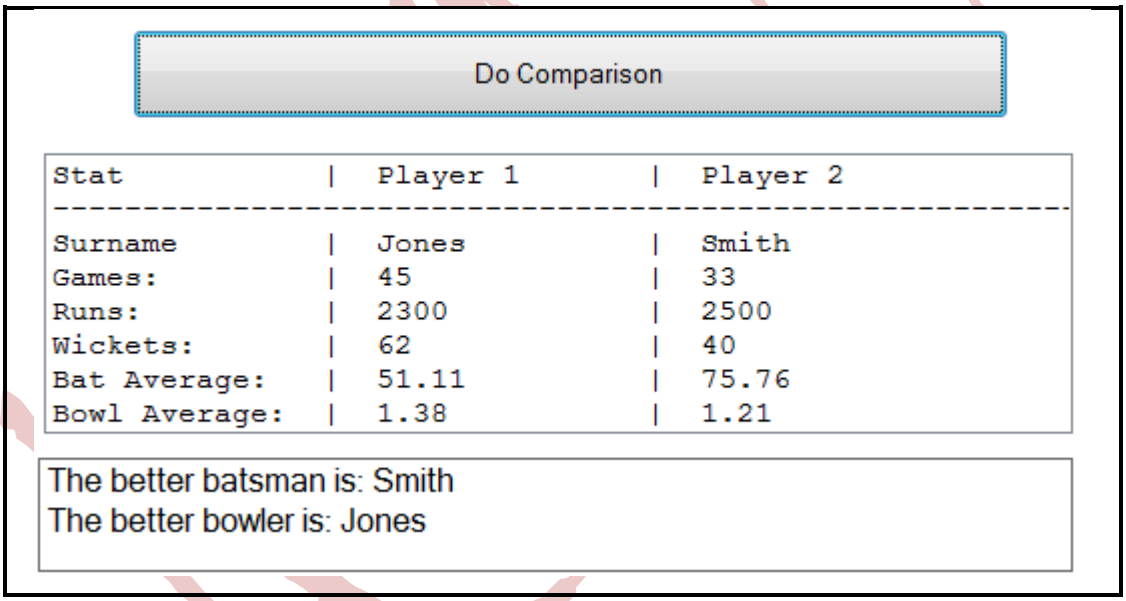
**STEP 1** RUN the program and enter all the *Player 1 Stats*:



**STEP 2** Enter all the *Player 2 Stats.*

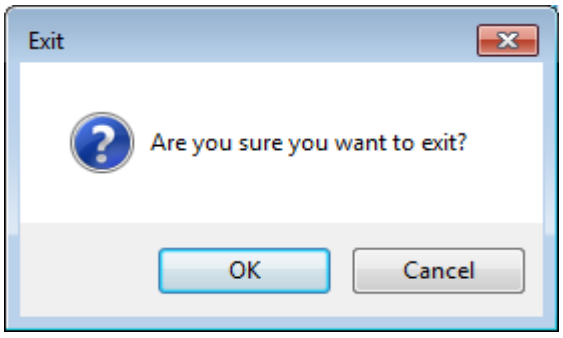



**STEP 3** Click on the *Do Comparison* button to display the output of the program in the list box and the display label:




**STEP 4** The *Clear* button resets and clears all the text boxes on the form, as well as the list box and label and then moves the focus to the *Player 1 Surname* text box:

**STEP 5** When the *Exit* button is clicked, the following message is displayed:




Clicking the *OK* button closes the application, while clicking the *Cancel* button leaves the program running.

Copyright reserved 

The application has already been designed and coded but it contains numerous syntax and logical errors, and you have been tasked with debugging the source code.

Below is a printout of the program code with all the syntax and logical errors still present:

```
…
Public Class Form1
Private Sub btnCompare_Click(ByVal sender As System.Object, ByVal e
As System.EventArgs) Handles btnCompare.Click
        'Player 1 variables
        Dim strP1Surname As String = 0
        Dim intP1Games As Integer = 0
        Dim intP1Runs As Integer = 0
        Dim intP1Wickets As Integer = 0
        Dim decP1BattingAverage As Decimal = 0
        Dim decP1BowlingAverage As Decimal = 0

        'Player 2 variables
        Dim strP2Surname As String = ""
        Dim decP2Games As Decimal = 0
        Dim intP2Runs As Integer = 0
        Dim intP2Wickets As Integer = 0
        Dim decP2BattingAverage As Decimal = 0
        Dim decP2BowlingAverage As Decimal = 0

        Declare Averages
        Dim strBetterBowler As String = ""
        Dim strBetterBatsman As String = ""

        strP1Surname = txtP1Surname.Text
        intP1Games = CInt(txtP1NoGames.Text)
        intP1Runs = CInt(txtP1Runs.Text)
        intP1Wickets = CInt(txtP1Wickets.Text)

        strP2Surname = txtP2Surname.Text
        intP2Games = CInt(txtP2NoGames.Text)
        intP2Runs = CInt(txtP2Runs.Text)
        intP2Wickets = CInt(txtP2Wickets)

        'Calculate Averages
        decP1BattingAverage = CDec(intP1Runs * intP1Games)
        decP1BowlingAverage = CDec(intP1Wickets / intP1Games)

        decP2BattingAverage = CDec(intP2Runs / intP2Games)
        decP2BowlingAverage = CDec(intP2Wickets / intP2Games)

        'Find better batsmen
        If decP1BattingAverage < decP2BattingAverage Then
            strBetterBatsman = strP1Surname
        ElseIf decP1BattingAverage < decP2BattingAverage Then
            strBetterBatsman = strP2Surname
        Else
            strBetterBatsman = "They are equal"
        End If
        'Find better bowler
        If decP1BowlingAverage > decP2BowlingAverage Then
            strBetterBowler = strP1Surname
        ElseIf decP1BowlingAverage < decP2BowlingAverage Then
            strBetterBowler = strP2Surname
        Else
            strBetterBowler = "They are equal"
        End If
```

Copyright reserved 

```
        'Display Comparisons
        lstDisplay.Items.Add("Stat".PadRight(15) & "|  " & "Player
1".PadRight(15) & "|  " & "Player 2")
        lstDisplay.Items.Add("-------------------------------------
-----------------")
        lstDisplay.Items.Add("Surname".PadRight(5) & "|  " &
strP1Surname.PadRight(15) & "|  " & strP2Surname)
        lstDisplay.Items.Add("Games:".PadRight(15) & "|  " &
intP1Games.ToString.PadRight(15) & "|  " & intP2Games.ToString)
        lstDisplay.Items.Add("Runs:".PadRight(15) & "|  " &
intP1Runs.ToString.PadRight(15) & "|  " & intP2Runs.ToString)
        lstDisplay.Items.Add("Wickets:".PadRight(15) & "|  " &
intP1Wickets.ToString.PadRight(15) & "|  " & intP2Wickets.ToString)
        lstDisplay.Items.Add("Bat Average:".PadRight(15) & "|  " &
decP1BattingAverage.ToString("F2").PadRight(15) & "|  " &
decP2BattingAverage.ToString("F2"))
        lstDisplay.Items.Add("Bowl Average:".PadRight(15) & "|  " &
decP1BowlingAverage.ToString("C2").PadRight(15) & "|  " &
decP2BowlingAverage.ToString("F2"))

        'Display better batsman and bowler
        lblDisplay.Text = "The better batsman is: " &
strBetterBatsman & ControlChars.NewLine & _
                          "The better bowler is: " & strBetterBowler

   Private Sub btnClear_Click(ByVal sender As System.Object, ByVal e
As System.EventArgs) Handles btnClear.Click
        'Clear textboxes
        txtP1NoGames.Clear()
        txtP1Runs.Clear()
        txtP1Surname.Clear()
        txtP1Wickets.Clear()

        txtP2NoGames.Clear()
        txtP2Runs.Clear()
        txtP2Surname.Clear()
        txtP2Wickets.Clear()

        'Clear label and listbox
        lstDisplay.clear()
        lblDisplay.Text = ""
    End Sub

   Private Sub btnExit_Click(ByVal sender As System.Object, ByVal e
As System.EventArgs) Handles btnExit.Click
        'Exit program
        Dim intAnswer As Integer
        intAnswer = MessageBox.Show("Are you sure you want to exit?",
"Exit", MessageBoxButtons.OKCancel, MessageBoxIcon.Question)
        If intAnswer = vbCancel Then
            Me.Close()
        End If
    End Sub
```

4.1 The completed program has been saved on your computer, but the program is not functioning properly.

Open the program ***PlayerComparison*** and correct all the *syntax errors* AND *logical errors* in the code. (18)

4.2 Add your EXAMINATION NUMBER to the source code as indicated and make a printout of the source code. (1)

4.3 Add you EXAMINATION NUMBER to the title bar of the form and run the program to compare the following two players:

| | Player 1 | Player 2 |
|---|---|---|
| Surname | Kallis | Ponting |
| Games | 57 | 33 |
| Runs | 2520 | 1215 |
| Wickets | 128 | 52 |

Make a printout of the form showing the correct result. (3)
**[22]**

**TOTAL: 100**

Copyright reserved